Week 1 — Task 2

Networking Basics for Defenders

Cyber Security Internship — Task Report

Status: Completed

Scope: OSI model, protocols, ports, DNS, and normal vs suspicious traffic

1. Introduction

Networking knowledge is a core requirement for defensive cybersecurity work. A SOC analyst needs to understand how systems communicate, which protocols are being used, what ports are exposed, and how normal traffic differs from potentially suspicious traffic.

This report documents the networking foundation requested in the internship task, together with the practical TryHackMe learning evidence supplied for the project.

2. Objectives

• Understand the OSI model from a defender's perspective.

• Understand common networking protocols and their purposes.

• Identify common ports used by network services.

• Understand how DNS converts domain names into IP addresses.

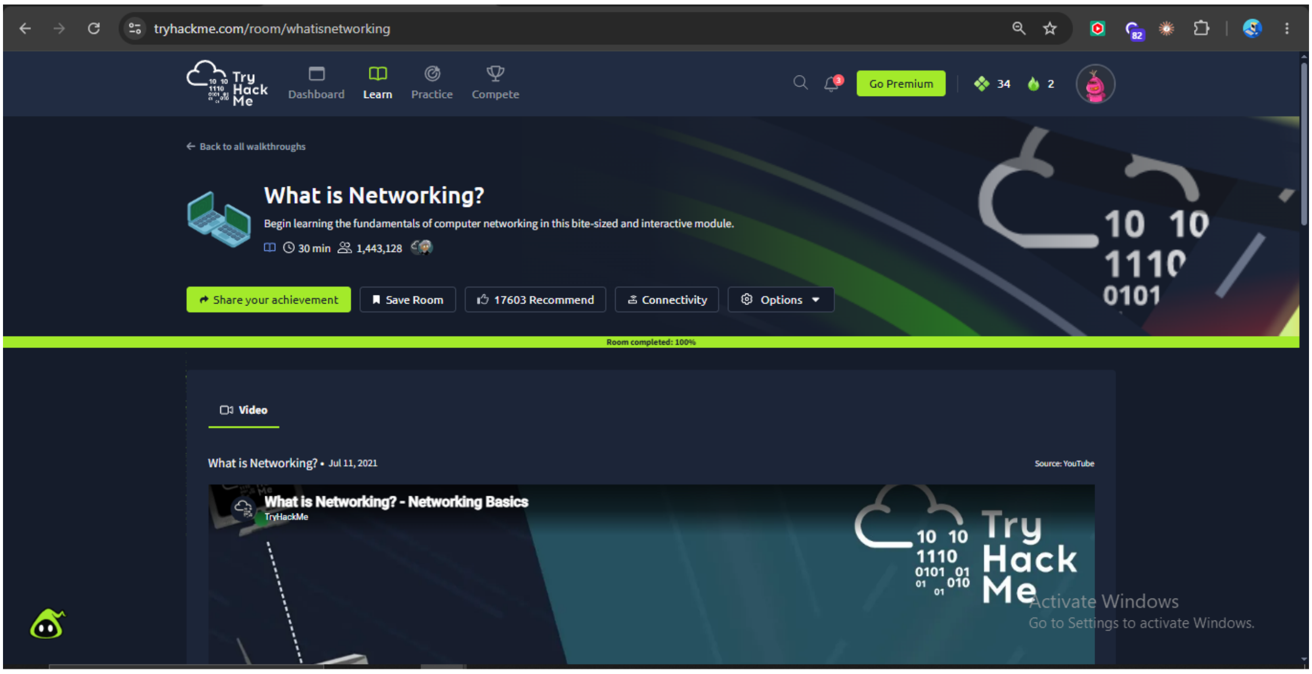
• Recognize an example of normal and suspicious traffic.

• Complete the assigned networking learning activities and document evidence.

3. Practical Learning Evidence

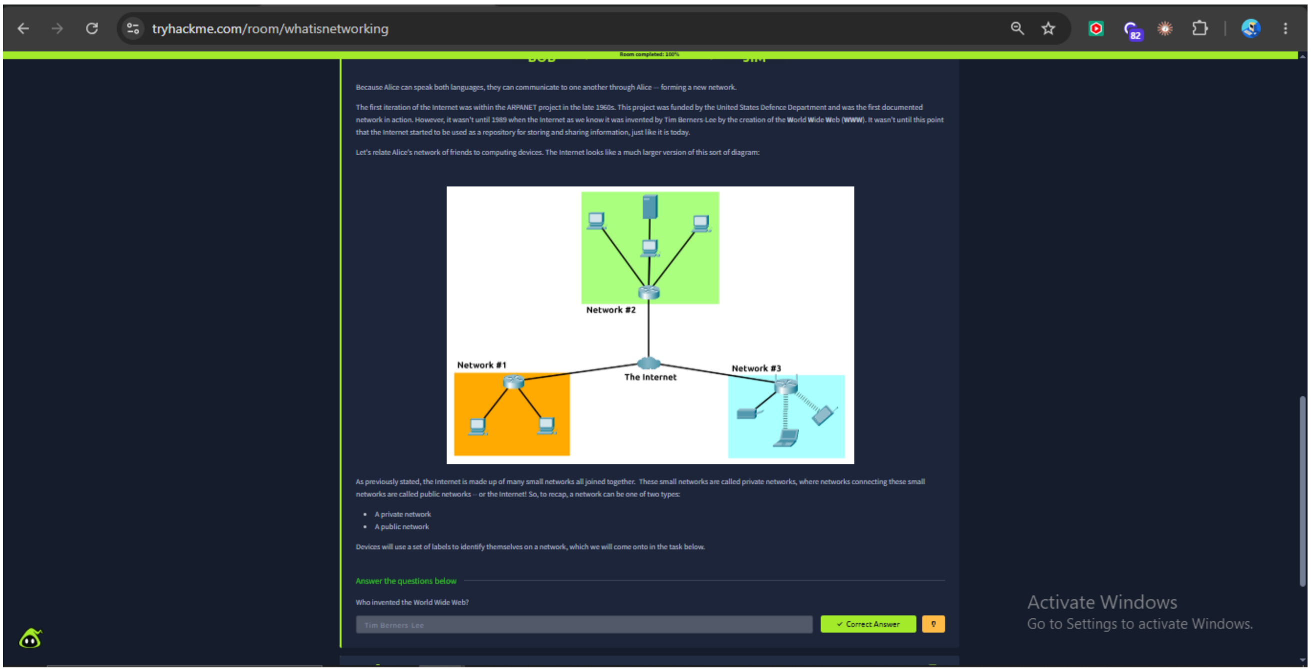
3.1 TryHackMe — What is Networking?

The assigned What is Networking? room was completed as part of the networking foundation.

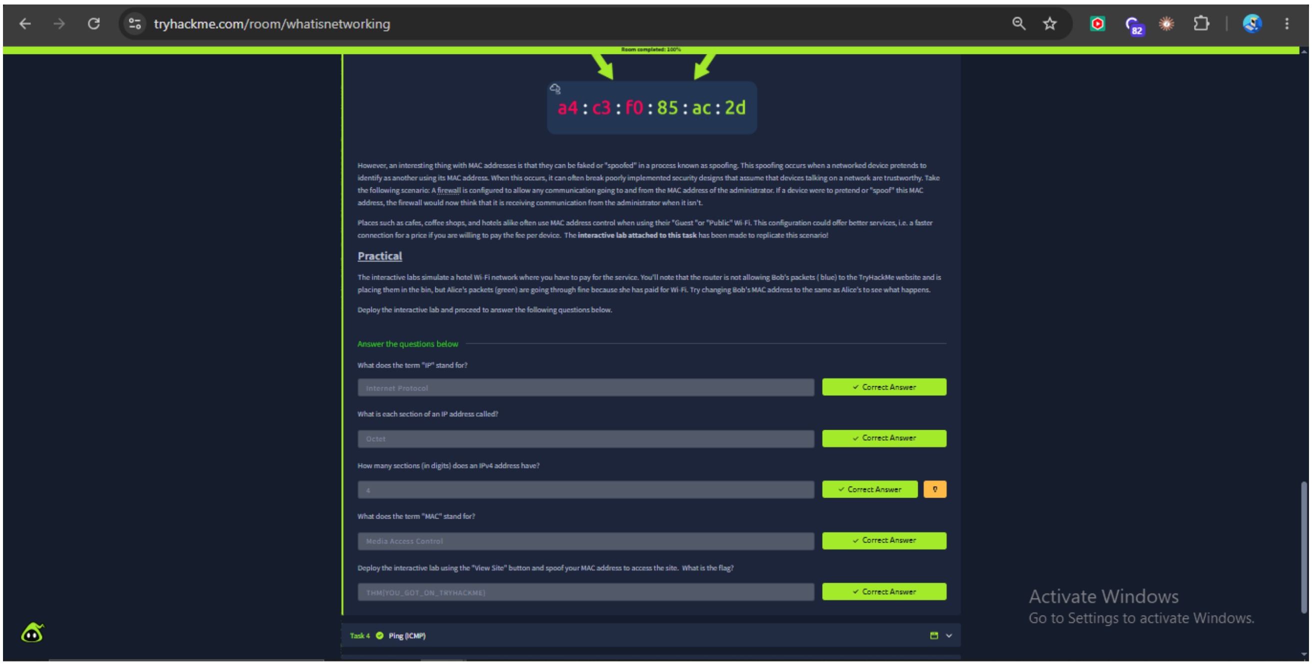


Original evidence screenshot: thm1.png

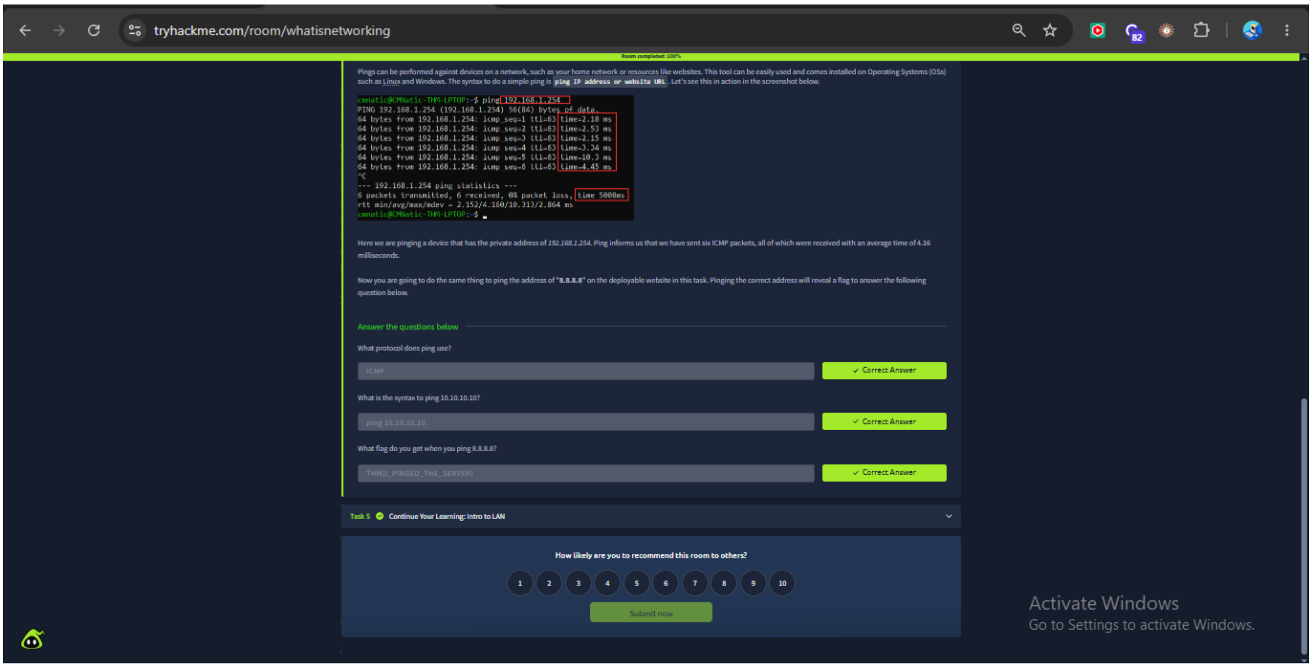
Original evidence screenshot: thm2.png



Original evidence screenshot: thm3.png



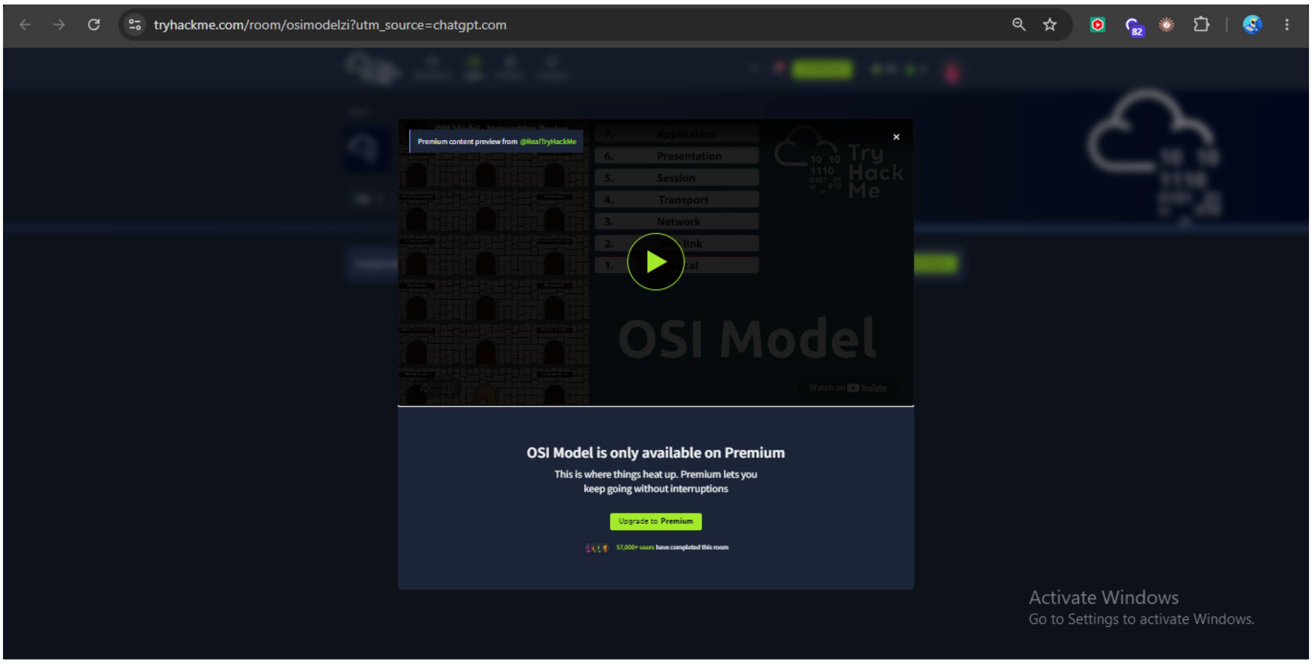
Original evidence screenshot: thm4.png



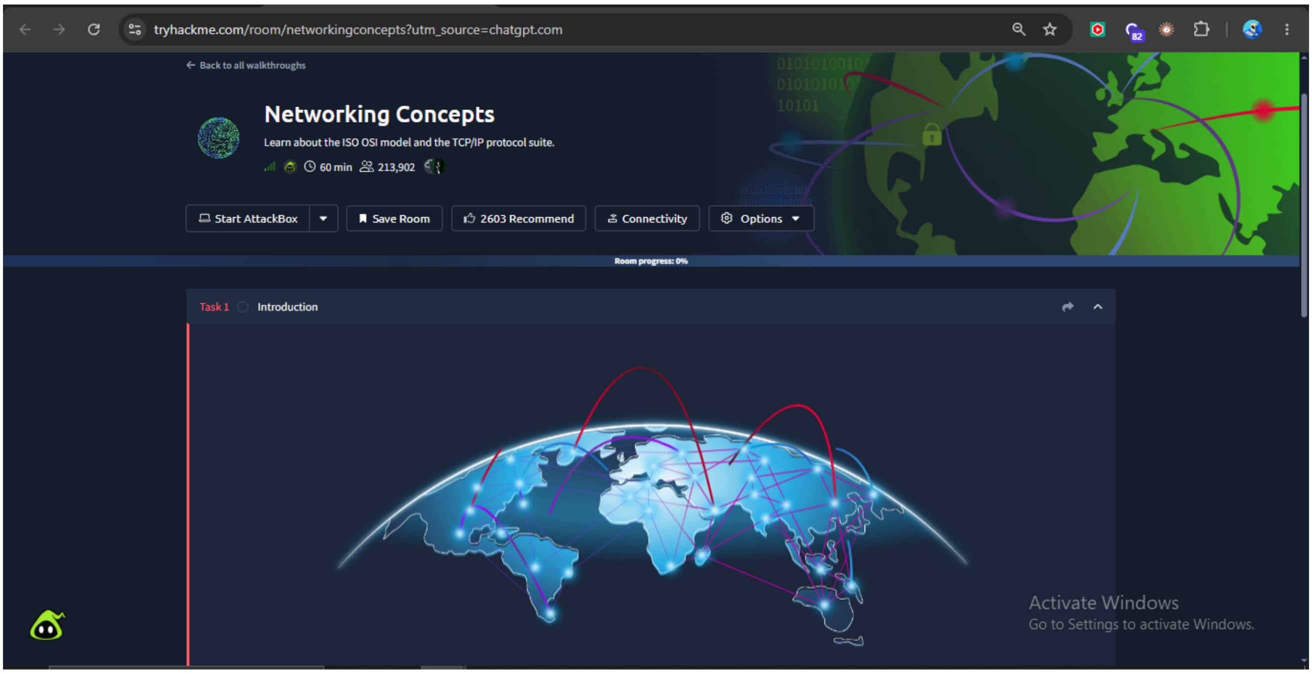
Original evidence screenshot: thm5.png

3.2 OSI Model Requirement — Free Alternative

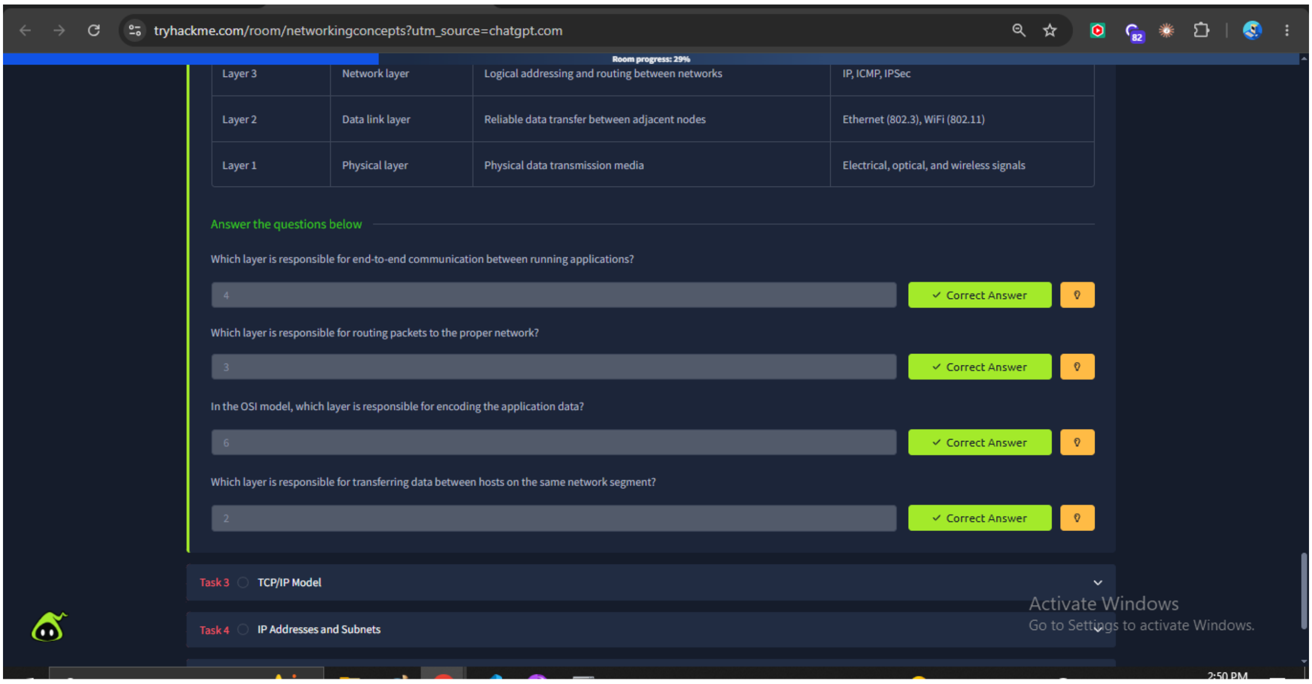
The originally listed OSI Model room displayed a Premium/subscription lock. The project therefore used the free Networking Concepts room to cover the OSI model requirement.



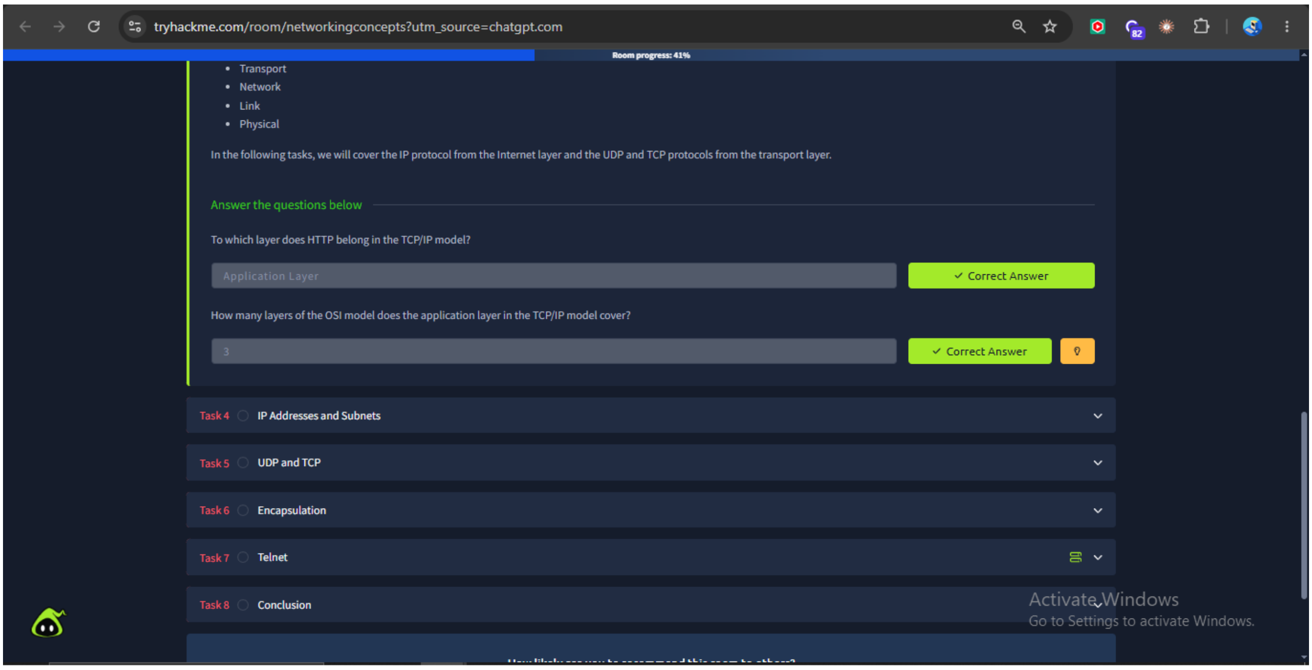
Original evidence screenshot: 2thm1.png — Premium access screen



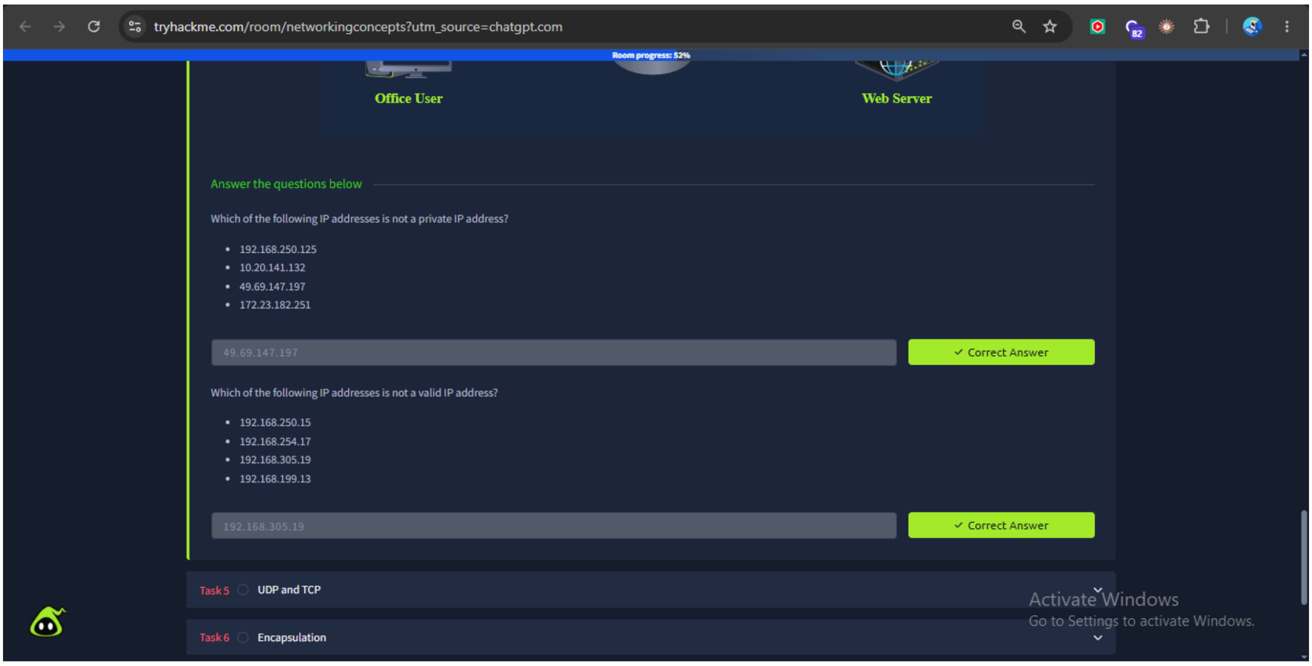
Original evidence screenshot: 2thm2.png



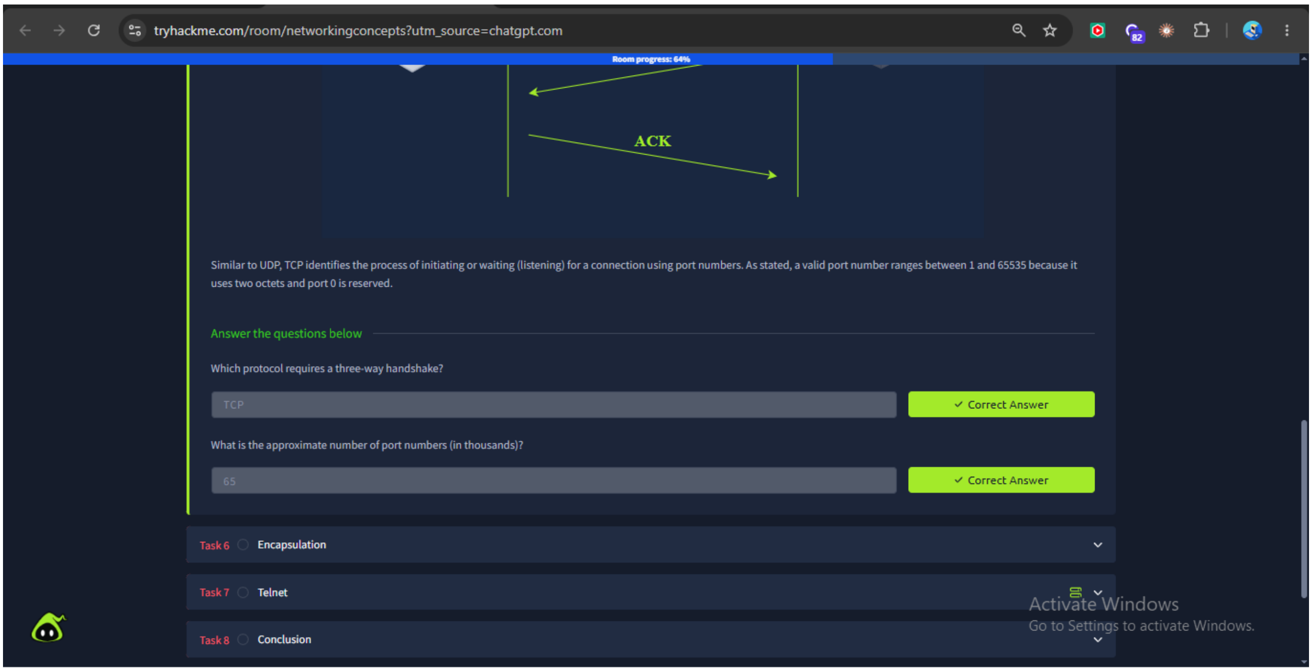
Original evidence screenshot: 2thm3.png



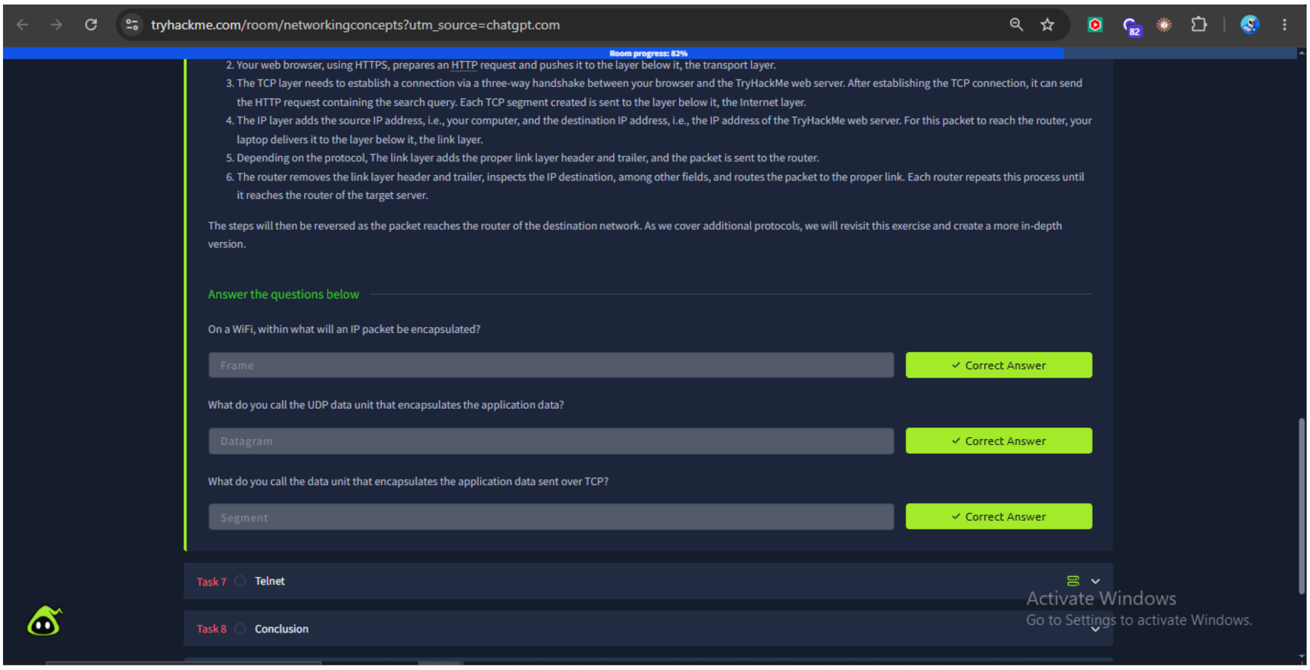
Original evidence screenshot: 2thm4.png



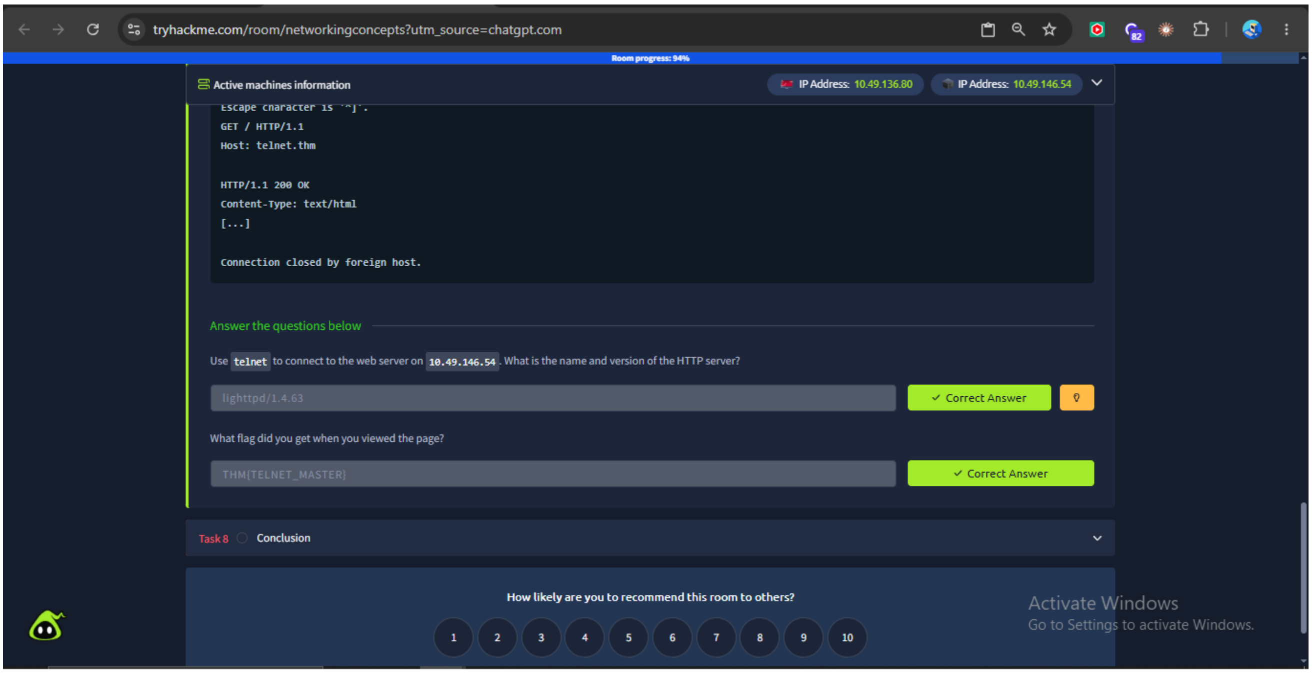
Original evidence screenshot: 2thm5.png



Original evidence screenshot: 2thm6.png



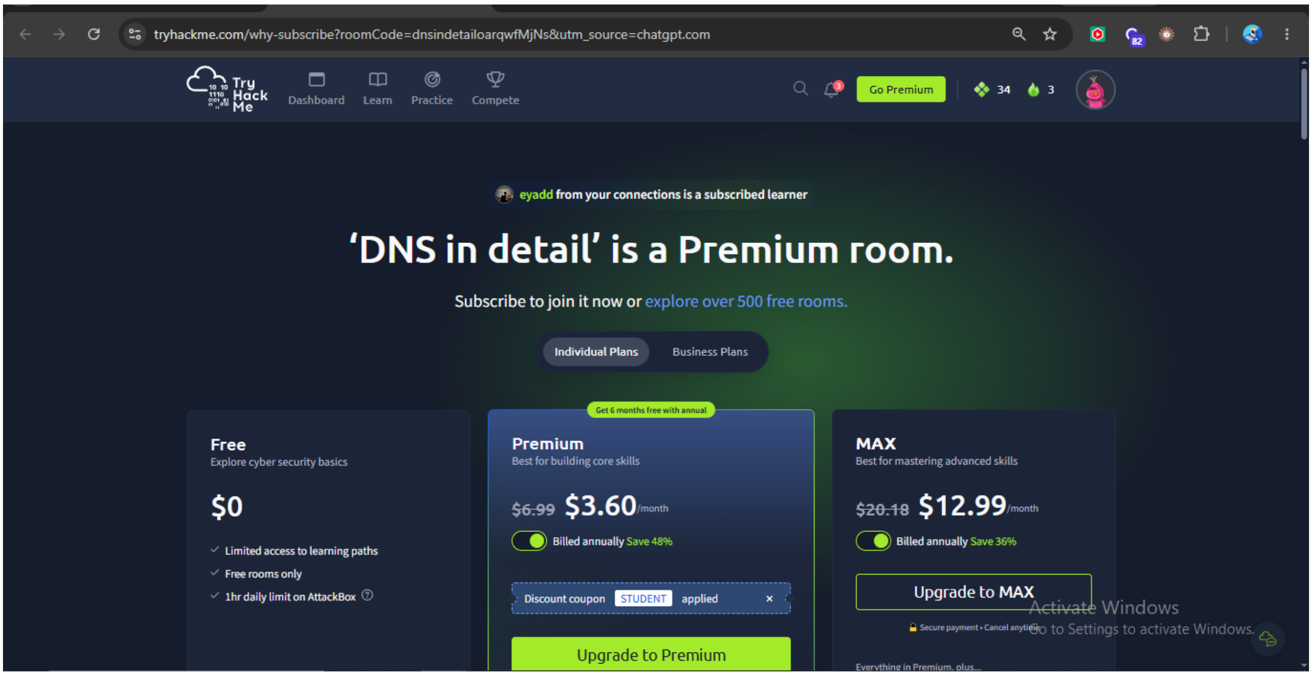
Original evidence screenshot: 2thm7.png



Original evidence screenshot: 2thm8.png

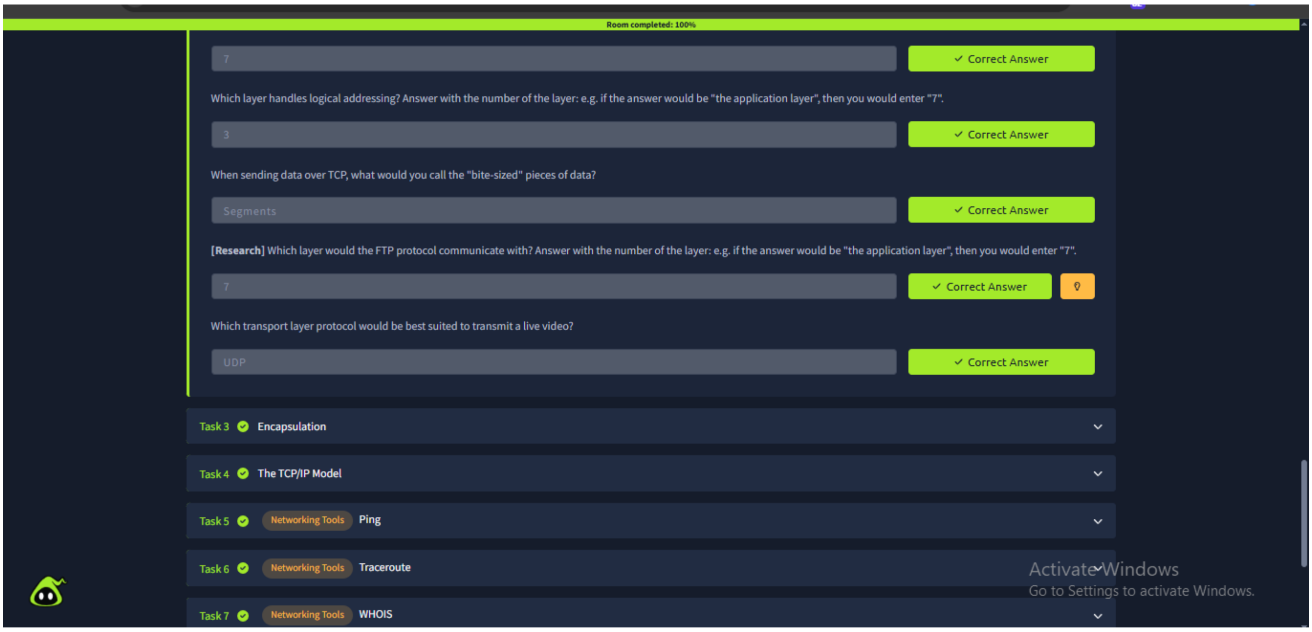
3.3 DNS Requirement — Free Alternative

The originally listed DNS in Detail room displayed a Premium/subscription lock. Introductory Networking was used as the free alternative for the DNS learning requirement.

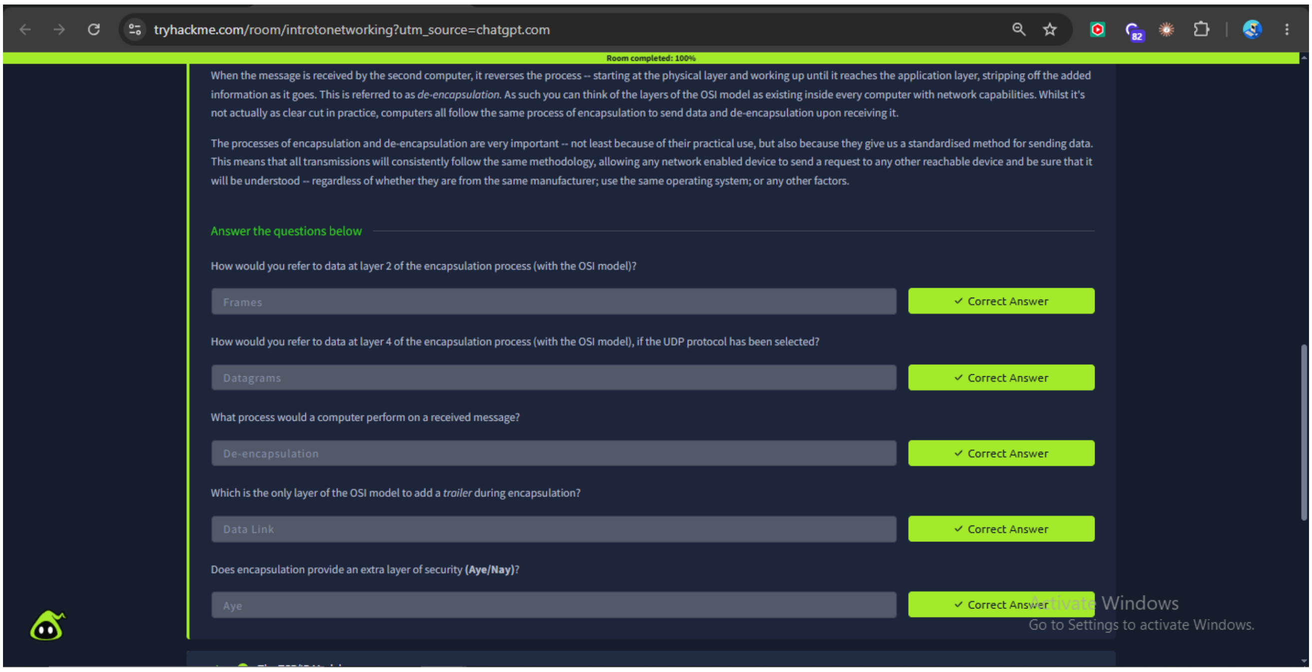


Original evidence screenshot: 3thm1.png — Premium access screen

Original evidence screenshot: 3thm2.png



Original evidence screenshot: 3thm3.png



Original evidence screenshot: 3thm4.png

Original evidence screenshot: 3thm5.png

4. OSI 7-Layer Model

The OSI (Open Systems Interconnection) model is a conceptual framework used to understand how network communication is organized into seven layers.

Layer	Name	Main Function	Examples
7	Application	Network services to applications	HTTP, HTTPS, DNS, SMTP, FTP, SSH
6	Presentation	Data representation, encoding, encryption, compression	JPEG, PNG, MIME, Unicode
5	Session	Establishes and manages sessions	RPC, session services
4	Transport	End-to-end communication and segmentation	TCP, UDP
3	Network	Logical addressing and routing	IP, ICMP, IPSec
2	Data Link	Local network communication	Ethernet, Wi-Fi
1	Physical	Transmission of raw bits	Cables, fiber, radio

Defender perspective: Layer 7 helps with application-level analysis such as HTTP and DNS; Layer 4 with TCP/UDP and ports; Layer 3 with IP addresses and routing; Layer 2 with local-network behavior; and Layer 1 with the physical/wireless medium.

5. Common Network Ports

Port	Service	Transport	Purpose	Defender Relevance
21	FTP	TCP	File transfer	Unencrypted traffic can expose data/credentials.
22	SSH	TCP	Secure remote administration	Watch for brute force and unusual remote logins.
23	Telnet	TCP	Remote administration	Normally insecure/unencrypted.
25	SMTP	TCP	Email transfer	Useful in mail-security investigations.
53	DNS	UDP/TCP	Domain resolution	Monitor suspicious domains and DNS patterns.
80	HTTP	TCP	Web traffic	Clear-text web traffic and malicious requests can be investigated.
110	POP3	TCP	Email retrieval	Older email protocol; unexpected use deserves review.
443	HTTPS	TCP	Encrypted web traffic	Normal service but also used by malicious sites/C2.
445	SMB	TCP	Windows file/printer sharing	High-value point for lateral-movement monitoring.
3389	RDP	TCP	Remote Desktop	Monitor exposed RDP and brute-force activity.

6. DNS Resolution Flow

DNS translates a human-readable domain name into an IP address that a computer can use for communication.

1	User enters a domain name
2	Local hosts file / local cache is checked
3	Recursive DNS resolver is queried
4	Resolver may query a root DNS server
5	Resolver queries the relevant TLD server
6	Resolver queries the authoritative DNS server
7	IP address is returned to the client
8	Client connects to the destination

DNS is important to defenders because malicious domains can support phishing, malware delivery, command-and-control communication, and other attacks. DNS logs can therefore provide useful investigation evidence.

7. Normal vs Suspicious Traffic

Normal example: A workstation resolves an expected company domain, connects to the returned destination over TCP/443, and loads a website. The destination, timing, and volume are consistent with normal user activity.

Suspicious example: A workstation repeatedly queries an unfamiliar external domain and then makes repeated outbound connections to an unusual destination. This should be investigated using DNS, network and endpoint telemetry.

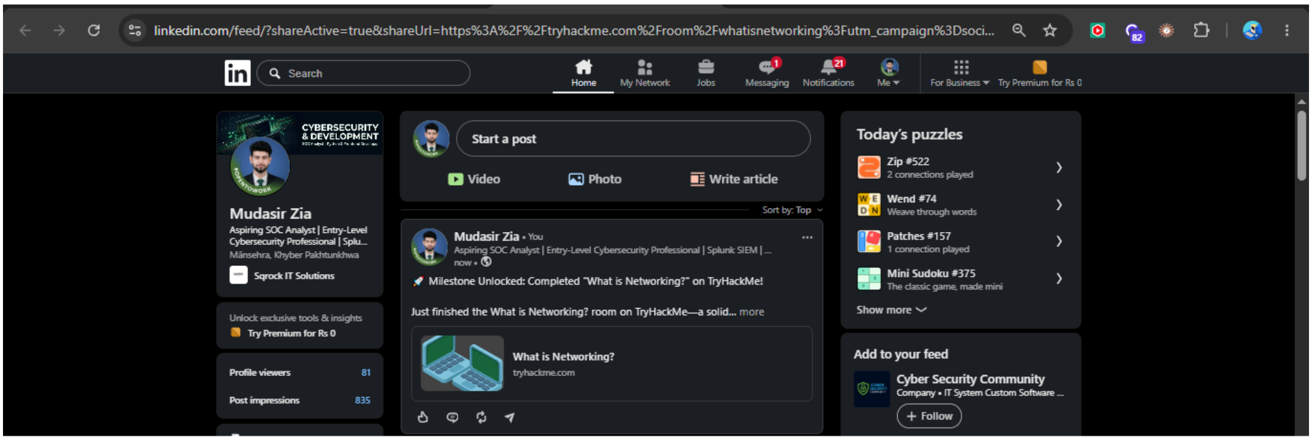
A SOC analyst should examine the source host, destination IP/domain, destination port, DNS queries, frequency of connections, and related endpoint activity before deciding whether the traffic is malicious.

8. Defender Takeaways

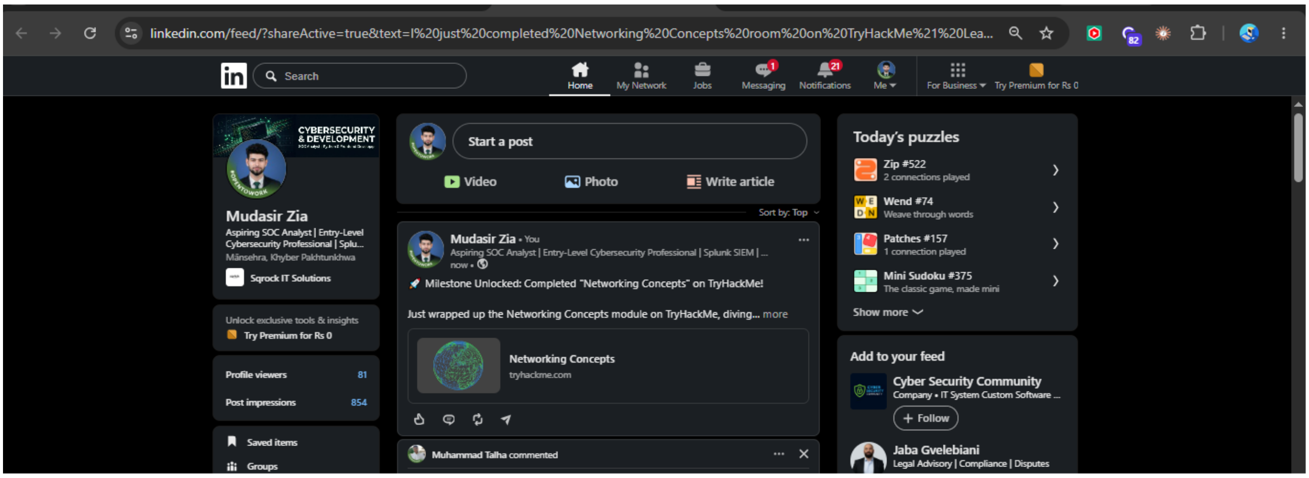
Networking information helps a SOC analyst answer practical investigation questions: Who communicated? Which service/port was involved? Which protocol was used? Which domain was resolved? Is the traffic expected? Does the connection pattern require further investigation?

9. Additional Supplied Evidence

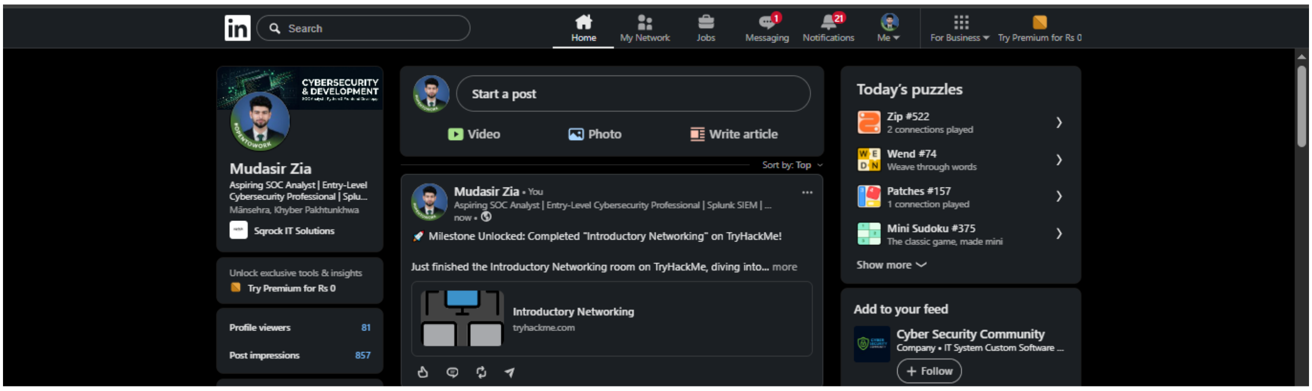
The uploaded evidence package also contained three LinkedIn screenshots. They are preserved below under their original filenames as supplied.



Original supplied screenshot: linkedin1.png



Original supplied screenshot: linkedin2.png



Original supplied screenshot: linkedin3.png

10. Conclusion

This task established a practical networking foundation for defensive cybersecurity work. The completed learning and analysis cover the OSI model, common protocols, important ports, DNS resolution, and the distinction between normal and suspicious traffic. These fundamentals are directly useful when interpreting firewall logs, packet captures, IDS/IPS alerts, DNS telemetry and endpoint security events.

Project Status: COMPLETED